

SPECIFICATION OF STATIONARY POINT PROCESSES BY PRODUCT-MOMENT DENSITIES

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SUMMARY

The concept of the covariance density, the covariance function of a stationary point process, is extended to product-moment densities of higher order. The relationship between the (central) product-moment densities and the 'renewal densities' is given. Sufficient conditions are stated for a stationary point process to be completely specified by the (central) product-moment densities or the 'renewal densities'. Further specifications are obtained for the subclasses of renewal processes and point processes with local aftereffects. A characterisation of the Poisson process in each of the classes considered is given.

1. Preliminaries and Definitions

The terminology in this paper is that of Steinijans (1970). In particular, Ω denotes the class of all countable subsets of the real line $\mathbb{R}$ that are finite in each finite interval. $\omega \in \Omega$ may be considered as a sequence of events occurring at the instants

$$T_i(\omega), i \in I := \{0, \pm 1, \pm 2, \dots\},$$

where the indices are allocated such that for $i, j \in I$ with $i < j$, $T_i(\omega) < T_j(\omega)$, with $T_0(\omega)$ denoting the largest non-positive value in the sequence.

$\mathcal{B}$ denotes the Borel- σ -algebra over $\mathbb{R}$, and $\overline{\mathcal{B}}$ the extended Borel- σ -algebra over the extended real numbers $\overline{\mathbb{R}}$. The functions

$$N(., \mathcal{B}): \Omega \rightarrow \overline{\mathbb{R}},$$

are for $\mathcal{B} \in \overline{\mathcal{B}}$ defined by

$$N(\omega, \mathcal{B}) := \text{card}(\omega \cap \mathcal{B}),$$

and

$$\mathcal{A} := \sigma\left(\bigcup_{\mathcal{B} \in \overline{\mathcal{B}}} N(., \mathcal{B})^{-1} \mathcal{B}\right)$$

denotes the induced σ -algebra.

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Let $P|A$ be a probability measure (for a specification of $P|A$ see e.g. Steinijs (1970)). Then we call the family of random variables $\{N(.,B), B \in \mathcal{B}\}$ the point process on $(\Omega, \mathcal{A}, P)$.

$\{N(.,B), B \in \mathcal{B}\}$ is called stationary iff for all $n \in \mathbb{N}$ ($\mathbb{N}$ denoting the set of natural numbers), all $B_i \in \mathcal{B}$, $i=1, \dots, n$ and all $t \in \mathbb{R}$ the two sets of random variables $N(., B_1), \dots, N(., B_n)$

and

$$N(., B_1^t), \dots, N(., B_n^t)$$

have the same joint distribution, where for $B \in \mathcal{B}$ and $r \in \mathbb{R}$ $r \in B^t$ iff $r-t \in B$.

In the case of stationarity we write independently of $s \in \mathbb{R}$

$$N(t) := N(\omega, (s, s+t]), \quad t \in [0, \infty).$$

Throughout this study we assume that $P\{\omega \in \Omega: N(\omega, \mathbb{R}) = 0\} = 0$, that $\nu := E[N(1)] < \infty$ (E standing for expectation (with respect to $P|A$ if not otherwise stated)) and that the probability of more than one event at a time t is zero. This means that the stationary point process under consideration is orderly with intensity $0 < \nu < \infty$.

The random variables

$$L_i(\omega) := T_{i+1}(\omega) - T_i(\omega), \quad i \in \mathbb{I},$$

defined on $(\Omega, \mathcal{A}, P)$ are called the lengths. Let Ω_0 denote the subspace of all sequences $\omega \in \Omega$ that contain an event at $t=0$, i.e.

$$\Omega_0 := \{\omega \in \Omega: \omega \cap \{0\} \neq \emptyset\},$$

and let $A_0 := A \cap \Omega_0$. Ryll-Nardzewski (1961) showed the existence of a conditional probability measure given an event at $t=0$, say $P_0|A_0$. The $A_0, \mathcal{B}$ -measurable lengths are called distances and are denoted by X_i , $i \in \mathbb{I}$. Intuitively speaking, 'lengths' corresponds to asynchronous sampling (from an arbitrary time instant) and 'distances' corresponds to synchronous sampling (from an arbitrary event).

For $n \in \mathbb{N}$ and $m_1, \dots, m_n \in \mathbb{N}$ let $f_{m_1, \dots, m_n}(t_1, \dots, t_n)$ denote the probability density function (p.d.f.) of the joint distribution of

$$\left(\sum_{i=1}^{m_1} X_i, \sum_{i=m_1+1}^{m_1+m_2} X_i, \dots, \sum_{i=m_1+\dots+m_{n-1}+1}^{m_1+\dots+m_n} X_i \right).$$

Then we call

$$(1) \quad h_n(t_1, \dots, t_n) := \sum_{m_1=1}^{\infty} \dots \sum_{m_n=1}^{\infty} f_{m_1, \dots, m_n}(t_1, \dots, t_n),$$

defined on $(0, \infty)^n$, the n -variate 'renewal density' (also 'renewal density' of order n). For a motivation of the name 'renewal density' see e.g. McFadden and Weissblum (1963).

2. (Central) Product-Moment Densities

For a stationary point process with intensity ν , $0 < \nu < \infty$, consider the process of difference quotients of the counting process, defined for $\Delta t > 0$ by

$$\Delta N(t) := \frac{1}{\Delta t} N(t, t + \Delta t).$$

Clearly $E[\Delta N(t)] = \nu$, independently of t and Δt .

Let $n \in \mathbb{N}$ and $t_i > 0$ ($i=1, \dots, n$). Then we call

$$\mu'_n(t_1, \dots, t_n) := \lim_{\Delta t \downarrow 0} \mu'_n(t_1, \dots, t_n; \Delta t),$$

where, setting $t_0 := 0$,

$$\mu'_n(t_1, \dots, t_n; \Delta t) := E\left[\frac{1}{\Delta t} \sum_{i=0}^n \Delta N\left(\sum_{j=0}^i t_j\right)\right],$$

the n -variate product-moment density (also p.m.d. of order n).

Analogously we call

$$\mu_n(t_1, \dots, t_n) := \lim_{\Delta t \downarrow 0} \mu_n(t_1, \dots, t_n; \Delta t),$$

where

$$\mu_n(t_1, \dots, t_n; \Delta t) := E\left[\frac{1}{\Delta t} \sum_{i=0}^n (\Delta N(\sum_{j=0}^i t_j) - \nu)\right],$$

the n -variate central product-moment density. Note that for $t > 0$ $\mu_1(t)$ is the covariance density introduced by Bartlett (1955).

In proposition 1 we relate the (central) product-moment densities to the 'renewal densities', thereby extending the well-known result (cf. e.g. Cox and Lewis (1966))

$$(2) \quad \mu_1(t) = \nu h_1(t) - \nu^2, \quad t > 0$$

to $n \geq 2$. In proposition 2 we show that (under suitable conditions) a stationary point process is specified by the (central) product-moment densities or by the 'renewal densities'. This generalises the result that a stationary renewal process is specified by the (univariate) renewal density or by the covariance density.

Proposition 1

Let $\{N(\cdot, B), B \in \mathcal{B}\}$ be a stationary and orderly point process with intensity ν , $0 < \nu < \infty$.

Let $n \in \mathbb{N}$. Suppose that for $i=1, \dots, n$ $h_i(t_1, \dots, t_i)$ is continuous in $(0, \infty)^i$. Then the (central) product-moment densities exist and are for $t_i > 0$ ($i=1, \dots, n$) given by

$$(3) \quad \mu'_n(t_1, \dots, t_n) = v h_n(t_1, \dots, t_n)$$

and

$$(4) \quad \mu_n(t_1, \dots, t_n) = v h_n(t_1, \dots, t_n)$$

$$\begin{aligned} & - v^2 (h_{n-1}(t_1, \dots, t_{n-1}) + h_{n-1}(t_2, \dots, t_n) + h_{n-1}(t_1 + t_2, t_3, \dots, t_n) \\ & + h_{n-1}(t_1, t_2 + t_3, t_4, \dots, t_n) + \dots + h_{n-1}(t_1, \dots, t_{n-2}, t_{n-1} + t_n)) \\ & + \dots \\ & + (-1)^{n-1} v^n \left(\sum_{i=1}^n h_1(t_i) + \sum_{i=1}^{n-1} h_1(t_i + t_{i+1}) + \dots + \sum_{i=1}^2 h_1(t_i + \dots + t_{i+n-2}) \right. \\ & \left. + h_1(t_1 + \dots + t_n) \right) \\ & + (-1)^n v^{n+1}, \end{aligned}$$

respectively. (Note that for $n=1$ (4) reduces to (2)).

Proof

Since $\{N(\cdot, B), B \in \mathcal{B}\}$ is orderly, $\mu_n(t_1, \dots, t_n; \Delta t)$ exists for sufficiently small $\Delta t > 0$ and is then for $0 < \Delta t < \min(t_1, \dots, t_n)$ given by

$$(5) \quad \mu'_n(t_1, \dots, t_n; \Delta t) = v \int_{-\Delta t}^{\Delta t} ds_1 \dots \int_{-\Delta t}^{\Delta t} ds_n h_n(t_1 - s_1, \dots, t_n - s_n) w_n(s_1, \dots, s_n; \Delta t),$$

where for $\Delta t > 0$ $\{w_n(\cdot; \Delta t); n \in \mathbb{N}\}$ is the class of consistent weighting functions which satisfies $w_n(\cdot; \Delta t) \geq 0$, $n \in \mathbb{N}$,

$$(6) \quad \int_{-\Delta t}^{\Delta t} dt_1 \dots \int_{-\Delta t}^{\Delta t} dt_n w_n(t_1, \dots, t_n; \Delta t) = 1, n \in \mathbb{N},$$

$$\begin{aligned} & \int_{-\Delta t}^{\Delta t} w_n(t_1, \dots, t_n; \Delta t) dt_j \\ & = w_{n-1}(t_1, \dots, t_{j-1}, t_{j+1}, \dots, t_n; \Delta t), j \in \{1, \dots, n\}, n \geq 2 \end{aligned}$$

with

$$w_1(t; \Delta t) = \frac{1}{(\Delta t)^2} (\Delta t - |t|) \quad \text{if } |t| \leq \Delta t,$$

0

elsewhere.

In order to prove (5) we write

$$\begin{aligned}\mu'_n(t_1, \dots, t_n; \Delta t) &= \frac{1}{(\Delta t)^{n+1}} E[N(o, \Delta t) N(t_1, t_1 + \Delta t) \dots N(t_1 + \dots + t_n, t_1 + \dots + t_n + \Delta t)] \\ &= \frac{1}{(\Delta t)^{n+1}} E \left[\int_{u=0}^{\Delta t} dN(u) \int_{u_1=(t_1-u)^+}^{t_1+\Delta t-u} dN(u+u_1) \dots \int_{u_n=(t_1+\dots+t_n+\Delta t-u-u_1-\dots-u_{n-1})^+}^{t_1+\dots+t_n+\Delta t-u-u_1-\dots-u_{n-1}} dN(u+u_1+\dots+u_n) \right]\end{aligned}$$

Interchanging the order of integration and expectation and observing that for $u_i > 0$ ($i=1, \dots, n$)

$$E[dN(u) dN(u+u_1) \dots dN(u+u_1+\dots+u_n)] = \nu h_n(u_1, \dots, u_n) du du_1 \dots du_n$$

(cf. e.g. Cox (1965)) yields*

$$(7) \quad \mu'_n(t_1, \dots, t_n; \Delta t) = \frac{1}{(\Delta t)^{n+1}} \int_0^{\Delta t} du \int_{t_1-u}^{t_1+\Delta t-u} du_1 \dots \int_{t_1+\dots+t_n-u-u_1-\dots-u_{n-1}}^{t_1+\dots+t_n+\Delta t-u-u_1-\dots-u_{n-1}} du_n h_n(u_1, \dots, u_n),$$

Interchanging the order of integration n -times yields after the substitution

$$s_i := t_i - u_i \quad (i=1, \dots, n) \quad (5).$$

The proof of proposition 1 is concluded by applying the second mean-value theorem for multiple integrals (cf. e.g. Apostol (1957)) to the r.h.s. of (5). By the postulated continuity of h_n and property (6) of w_n this implies (3). (4) then is an immediate consequence of (3) and the relationship between the central moments $\mu_n(t_1, \dots, t_n; \Delta t)$ and the moments $\mu'_n(t_1, \dots, t_n; \Delta t)$.

*Another proof of (7) runs as follows (cf. Steinijs (1971)). Since the counting process $\{N(t), t \in [0, \infty)\}$ is the inverse process of the process of waiting times $\{T_n(\cdot), n \in \mathbb{N}\}$, it is convenient to express the moments of $N(t)$ in terms of the distribution functions of the $T_n(\cdot)$. The p.d.f.'s of the $T_n(\cdot)$, which are based on lengths, are then expressed in terms of the p.d.f.'s of the corresponding sums of consecutive distances. This is best done by means of the respective Laplace transforms (cf. e.g. McFadden and Weissblum (1963)). The proof therefore makes extensive use of multiple Laplace transforms.

Proposition 2

Let $\{N(\cdot, B), B \in \mathcal{B}\}$ be a stationary and orderly point process with intensity ν , $0 < \nu < \infty$.

Suppose that for $n \in \mathbb{N}$ the n -variate 'renewal density' is continuous in $(0, \infty)^n$ and that

$$h_n(0, \dots, 0) := \lim_{\Delta t_1 \downarrow 0} \dots \lim_{\Delta t_n \downarrow 0} h_n(\Delta t_1, \dots, \Delta t_n)$$

exists. Furthermore, suppose that for all

$$0 \leq t_1 < t_1 + \Delta t_1 \leq t_2 < t_2 + \Delta t_2 \leq \dots \leq t_n < t_n + \Delta t_n, \quad n \in \mathbb{N},$$

the moments

$$(8) \quad \mu'_{k_1, \dots, k_n}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n) := E[N(t_1, t_1 + \Delta t_1)^{k_1} \dots N(t_n, t_n + \Delta t_n)^{k_n}],$$

$$k_i = 0, 1, \dots \quad (i = 1, \dots, n)$$

satisfy

$$(9) \quad \sum_{m=1}^{\infty} \lambda_{2m}^{-1/2m}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n) = \infty,$$

where

$$(10) \quad \lambda_{2m}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n)$$

$$:= \mu'_{2m, 0, \dots, 0}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n) + \dots +$$

$$\mu'_{0, \dots, 0, 2m}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n).$$

Then the stationary point process $\{N(\cdot, B), B \in \mathcal{B}\}$ is uniquely specified by the (central) product-moment densities or by the 'renewal densities'.

Proof:

Similarly to (5) one can show that under the assumed continuity of h_n in $[0, \infty)^n$ the moments (8) can be expressed in terms of the 'renewal densities'.

Under the assumption (9) the moments (8) in turn uniquely determine for $n \in \mathbb{N}$ the n -dimensional distribution function of the joint distribution of $N(t_1, t_1 + \Delta t_1), \dots, N(t_n, t_n + \Delta t_n)$. This follows from a well-known theorem on the n -dimensional moment problem (cf. e.g. Shohat and Tamarkin (1950)).

Finally, a stationary point process is uniquely specified by the (consistent and stationary) distribution of counts in any finite number of disjoint, finite, left open and right closed intervals in $\mathbb{R}$ (cf. e.g. Steinijans (1970)).

Corollary 1

For the Poisson process all central product-moment densities are identically zero, i.e.

$$(11) \quad \mu_n(u_1, \dots, u_n) = 0 \text{ for all } u_i \geq 0 \ (i=1, \dots, n), n \in \mathbb{N}.$$

Moreover, the Poisson process is the only stationary point process with this property.

Proof:

For a Poisson process with intensity ν , $0 < \nu < \infty$, it follows that for $u_i \geq 0$ ($i=1, \dots, n$)

$$\begin{aligned} f_{m_1, \dots, m_n}(u_1, \dots, u_n) \\ = \nu^{m_1 + \dots + m_n} e^{-\nu(u_1 + \dots + u_n)} \\ \frac{u_1^{m_1-1}}{(m_1-1)!} \frac{u_2^{m_2-1}}{(m_2-1)!} \dots \frac{u_n^{m_n-1}}{(m_n-1)!} \end{aligned}$$

and hence

$$(12) \quad h_n(u_1, \dots, u_n) = \nu^n \text{ for all } u_i \geq 0 \ (i=1, \dots, n), n \in \mathbb{N}.$$

By (4) this is obviously equivalent to (11), which proves the first part of the assertion.

Conversely, we have to show that a stationary point process which satisfies (11) and hence (12) is the Poisson process. By proposition 2 it is sufficient to show that the moments (8), which are the same for the two processes under consideration, satisfy (9). This can be seen as follows. By definition (10) it follows that

$$\lambda_{2m}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n) = \sum_{i=1}^n E[N(t_i; t_i + \Delta t_i)^{2m}] \leq n E[N(t)^{2m}],$$

where $t := \max(\Delta t_1, \dots, \Delta t_n)$. For a Poisson process with intensity ν

$$\mu'_k(0; t) := E[N(t)^k]$$

satisfies the recursion formula

$$\mu'_{k+1}(0; t) = \nu t \sum_{j=0}^k \binom{k}{j} \mu'_{k-j}(0; t)$$

(cf. e.g. Haight (1967)). It is then easily shown by induction over k that

$$\mu'_k(0;t) \leq (\nu t + k)^k, \quad k \in \mathbb{N},$$

and hence

$$\lambda_{2m}(t_1, \dots, t_n; \Delta t_1, \dots, \Delta t_n) \leq n(\nu t + 2m)^{2m}.$$

As for all $\nu > 0$, $t > 0$ the sequences $((n(\nu t + 2m)^{2m})^{-1/2m})_{m \in \mathbb{N}}$ and $((2m)^{-1})_{m \in \mathbb{N}}$ are asymptotically equal, (9) follows from the divergence of the harmonic series.

3. Renewal Processes

A point process is called a renewal process if the distances X_i , $i \in \mathbb{I}$ are independently identically distributed. A renewal process is called stationary (equilibrium) iff the forward recurrence time $T_1(\cdot)$ has p.d.f.

$$f_{T_1}(t) = \begin{cases} \nu \int_t^\infty f_X(u) du, & t > 0, \\ 0 & \text{elsewhere,} \end{cases}$$

where f_X is the p.d.f. of the common distribution of the distances and $\nu = (E_{P_0}[X])^{-1}$ is the intensity of the process; it is called ordinary iff

$$f_{T_1}(t) = f_X(t), \quad t \in \mathbb{R}.$$

Since for a renewal process the distances are independent, each term on the r.h.s. of (1) factorises, so that for $n \in \mathbb{N}$

$$h_n(t_1, \dots, t_n) = h_1(t_1) \dots h_1(t_n), \quad t_i > 0 \quad (i=1, \dots, n).$$

It then follows from proposition 2 that a renewal process is completely specified by the renewal density h_1 and hence by the univariate product-moment density μ'_1 or by the covariance density μ_1 . This of course can be seen directly from (14) below.

Proposition 3

Under the assumption that $h_1(t)$ is continuous in $(0, \infty)$, a stationary (and orderly) renewal process is the Poisson process iff the covariance density $\mu_1(t)$ is identically zero for $t > 0$.

Proof:

It is easily shown that a renewal process is the Poisson process iff it is ordinary and stationary, and then

$$(13) \quad f_{T_1}(t) = f_X(t) = \nu e^{-\nu t}, t \geq 0.$$

By proposition 1 the assertion is equivalent to

$$h_1(t) = \nu, t \geq 0,$$

which in turn is equivalent to (13). This follows immediately from the well-known relation

$$(14) \quad h_1^*(s) = \frac{f_X^*(s)}{1 - f_X^*(s)}$$

(cf. e.g. Cox and Lewis (1966)) between the Laplace transforms of h_1 and f_X .

It should be emphasised that the given characterisation of the Poisson process only holds in the class of stationary renewal processes. To show this we refer to theorem 1 by Störmer (1969) which states that to each renewal process defined by the distances $\{X_0, X_1, X_2, \dots\}$ there exists a point process which is not a renewal process but has the same distribution of the number of points in $(0, t]$ as the original process.

The theorem is proved by constructing a random variable $\bar{X}_2$ which depends on X_1 such that the point process defined by $\{X_0, X_1, \bar{X}_2, X_3, \dots\}$ has the desired property.

In particular, it follows that a point process satisfying

$$P_0(S_n \leq t) = P_0(\bar{S}_n \leq t), t \geq 0,$$

where S_n denotes the sum of the first n distances in a Poisson process and $\bar{S}_n$ is obtained from it by replacing X_2 by $\bar{X}_2$, is not a Poisson process but has zero covariance density.

4. Processes with Local Aftereffects

A stationary point process is said to have local aftereffects over k steps (to be of local contagion over k steps (cf. e.g. Wold (1948))) if the stationary sequence of distances forms a (generalised) multiple Markov chain of order k , i.e. if there exists an index $k \in \mathbb{N}$ such that for all $j \geq k$ and all $x_j \geq 0$ ($i \in I$), $n \in I$

$$(15) \quad P_0(X_n \leq x_n \mid X_{n-j} = x_{n-j}, \dots, X_{n-1} = x_{n-1}) \\ = P_0(X_n \leq x_n \mid X_{n-k} = x_{n-k}, \dots, X_{n-1} = x_{n-1}).$$

(Here and in the following we assume the existence of the conditional probabilities in question.) Clearly, a stationary point process which has local aftereffects over k steps, has local aftereffects over j steps for all $j \geq k$, so that in particular a renewal process has local aftereffects over k steps for all $k \in \mathbb{N}$ although no genuine aftereffects (in the above defined sense) are present.

Our aim is to show that a stationary point process with local aftereffects over $k \geq 1$ steps is already specified by the (central) product-moment densities of up to order $k+1$ or by the corresponding 'renewal densities'.

Proposition 4

Let $\{N(\cdot, B), B \in \mathcal{B}\}$ be a stationary point process with local aftereffects over $k \geq 1$ steps which satisfies the conditions of proposition 2. Under suitable regularity conditions for the existence of the conditional probabilities in question it follows that for $n > k+1$, $t_i > 0$ ($i=1, \dots, n$), $n \in \mathbb{N}$

$$(16) \quad h_n(t_1, \dots, t_n) = h_{k+1}(t_1, \dots, t_{k+1}) \frac{h_{k+1}(t_2, \dots, t_{k+2})}{h_k(t_2, \dots, t_{k+1})} \dots \frac{h_{k+1}(t_{n-k}, \dots, t_n)}{h_k(t_{n-k}, \dots, t_{n-1})},$$

and hence that $\{N(\cdot, B), B \in \mathcal{B}\}$ is already specified by $\{h_n(\cdot), n=1, \dots, k+1\}$

Proof:

Definition (1) of the 'renewal densities' leads for sufficiently small Δt , $0 < \Delta t < \min(t_1, \dots, t_n)$, to the interpretation of $h_n(t_1, \dots, t_n) (\Delta t)^n$ as the conditional probability of an event in each of the intervals $(t_1 + \dots + t_i, t_1 + \dots + t_i + \Delta t]$, $i=1, \dots, n$, given an event at $t=0$. This probability is also given by

$$(17) \quad P_0(A_1 A_2 \dots A_n) = P_0(A_1) P_0(A_2|A_1) P(A_3|A_1 A_2) \dots P_0(A_n|A_1 \dots A_{n-1}),$$

where for $i \in \mathbb{N}$

$$A_i := \{\omega \in \Omega_0 : \omega \cap (t_1 + \dots + t_i, t_1 + \dots + t_i + \Delta t] \neq \emptyset\}.$$

Since any two points of $\omega \in \Omega_0$ are separated by at least one distance, it follows from (15) that for $j > k$

$$P_0(A_j|A_1 \dots A_{j-1}) = P_0(A_j|A_{j-k}, \dots, A_{j-1}) = \frac{P_0(A_{j-k} \dots A_{j-1} A_j)}{P_0(A_{j-k} \dots A_{j-1})}.$$

This in turn corresponds for sufficiently small $\Delta t > 0$ to

$$\frac{h_{k+1}(t_{j-k}, \dots, t_j) (\Delta t)^{k+1}}{h_k(t_{j-k}, \dots, t_{j-1}) (\Delta t)^k},$$

and (16) follows from (17).

Corollary 2

Under the assumptions of proposition 4 it follows that the Poisson process is the only stationary point process with local aftereffects over k steps, $k \in \mathbb{N}$, for which all central product-moment densities of up to order $k+1$ are identically zero, i.e.

$$(18) \quad \mu_i(t_1, \dots, t_i) = 0, \quad t_j > 0 \quad (j=1, \dots, i) \text{ for } i=1, \dots, k+1.$$

Proof:

By corollary 1 we only have to show that such a process satisfies (12) for all $n \in \mathbb{N}$. This follows from (18) for $n \leq k+1$ by (4), and then for $n > k+1$ from (16).

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